

Linux Security Monitoring and Auditing

WEEK 4 Practical Lab

Student Name: Faith Mathapelo Letsoara

Registration number: C11/26/CGRC/17582

Cohort /Batch: Cohort 11

Instructor Name: Steve Wamala Gamba

Submission Date :08/10/2026

Executive Summary

The purpose of this assessment was to evaluate monitoring, auditing and security-hardening controls in an authorised Kali Linux virtual machine. The assessment focused on Linux audit configuration, security event generation, log analysis, Lynis hardening assessment, governance mapping and retesting activities.

Auditd, ausearch, aureport and journalctl were used to verify that security-relevant activity could be recorded, retrieved and reviewed. Custom audit rules monitored changes to sensitive account files and program execution, while system journal evidence was reviewed for privileged activity, authentication failures, warnings and errors.

A Lynis security assessment was completed using Lynis version 3.1.6. The system produced a Hardening Index of 61. Three findings were selected for risk interpretation: missing PAM password-strength testing (AUTH-9262), password-protected accounts without expiry dates (AUTH-9282), and enabled USB-storage support where removable storage might not be required (USB-1000).

Overall, the assessment confirmed that audit and log monitoring were operational in the lab. The Lynis results identified clear hardening opportunities that should be assigned to control owners, remediated according to approved policy, and retested to confirm improvement.

1. Scope and Authorization

Scope

The assessment was performed on an authorised Kali Linux VirtualBox virtual machine named faithlets-kali and included Auditd verification, custom audit rules, event generation, Linux log analysis, Lynis hardening assessment, governance mapping, remediation planning and retesting recommendations.

- Verification of Auditd installation and configuration
- Creation and review of audit monitoring rules
- Security event generation
- Linux log analysis
- Security assessment using Lynis
- Governance and control monitoring review
- Remediation planning and retesting recommendations

Authorization

All activities were performed in the authorised isolated lab environment and were limited to approved educational exercises. No production or third-party systems were targeted.

2. Methodology

The assessment followed a structured approach consisting of four phases.

Phase 1 – Audit Configuration and Verification

Auditd and supporting audit tools were verified to determine whether security event monitoring capabilities were available.

Commands reviewed included:

auditctl
ausearch
aureport

Custom audit rules were created for:

- Monitoring changes to /etc/passwd
- Monitoring changes to /etc/shadow
- Monitoring program execution events
- Monitoring authentication-related activity

Phase 2 – Security Event Generation

Test activities were performed to generate auditable events and validate monitoring controls.

Examples included:

nano /etc/passwd
ls /tmp

These actions were intended to trigger configured audit rules and generate audit records.

Phase 3 – Linux Log Analysis

Linux logging mechanisms were reviewed using:
journalctl

The /var/log directory and related log locations were examined to verify the availability of security logging data.

Phase 4 – Security Assessment

The Lynis security auditing tool was installed and executed to assess system security posture and hardening effectiveness.

sudo lynis audit system

3. Evidence Bundle 1 – Auditd Configuration and Events

Objective

To verify that Linux auditing tools were installed and capable of supporting security monitoring activities.

Observations

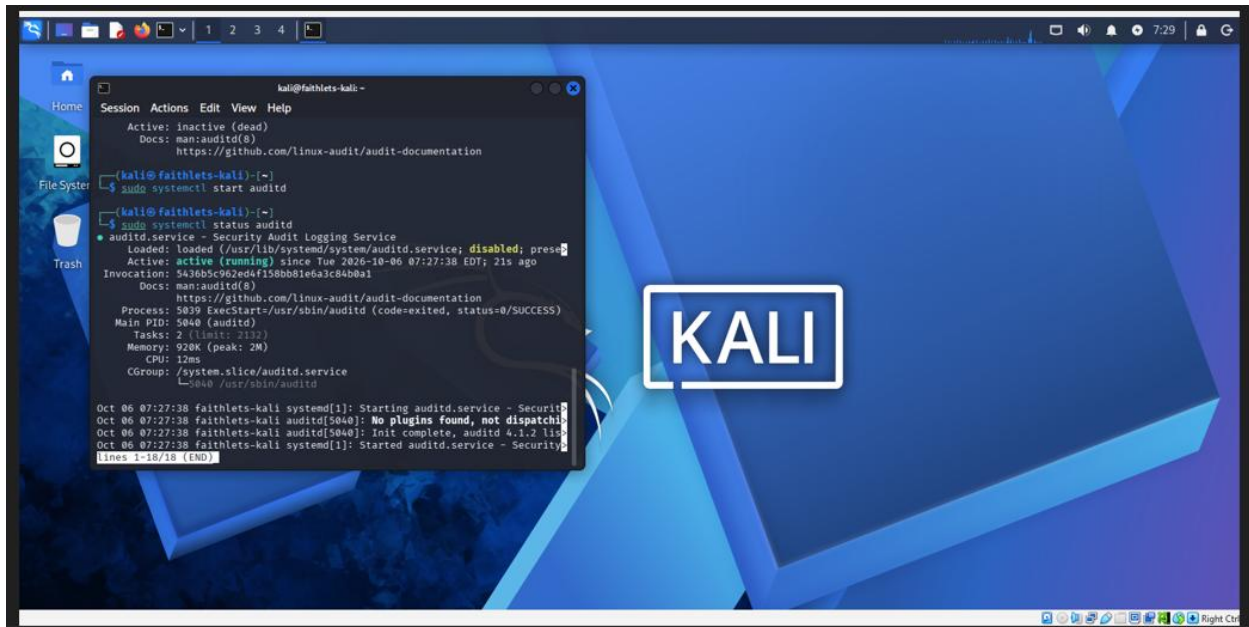


Figure 1: Auditd service active and running

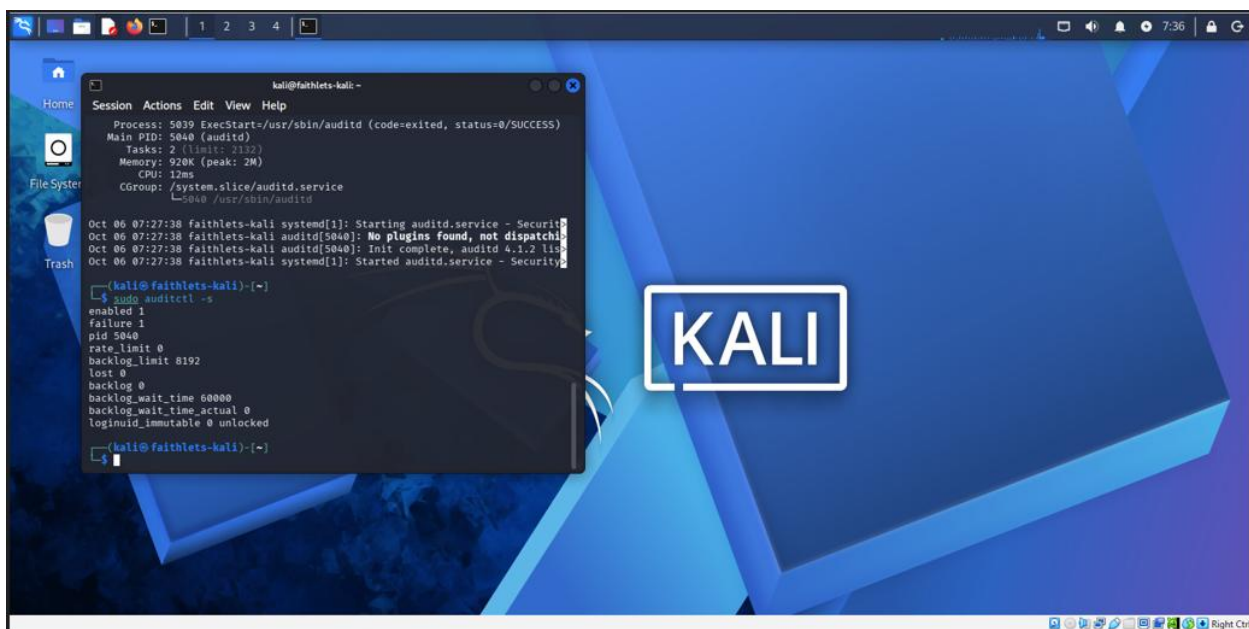


Figure 2: Audit subsystem status showing enabled=1

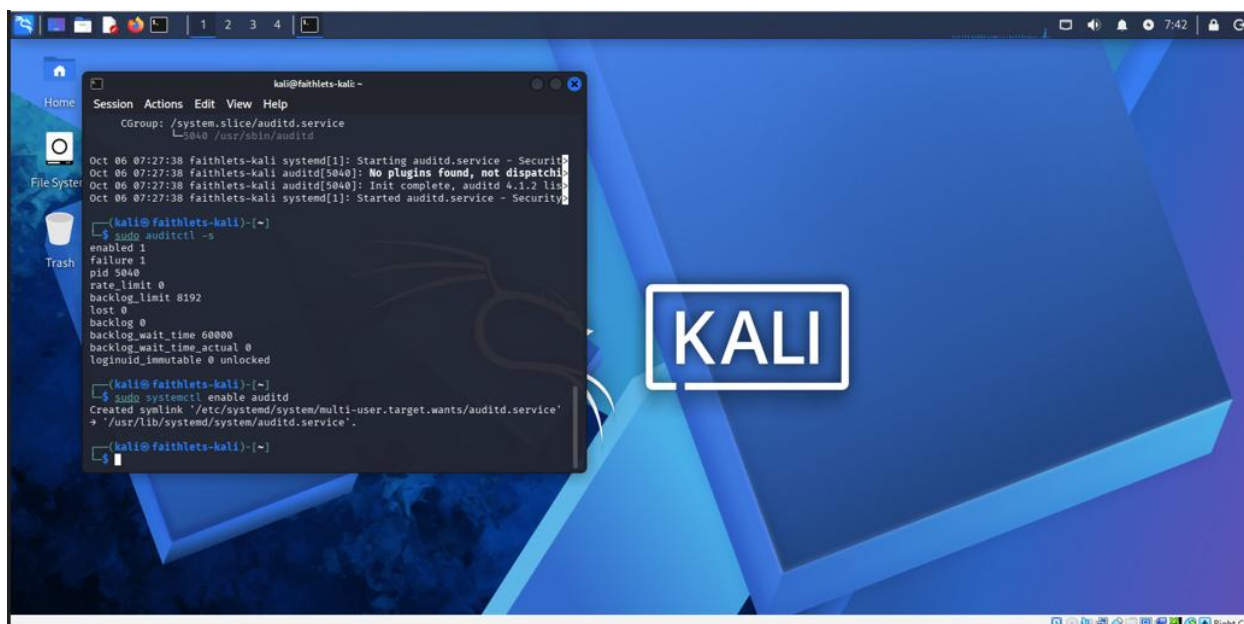


Figure 3: Auditd enabled to start automatically at boot

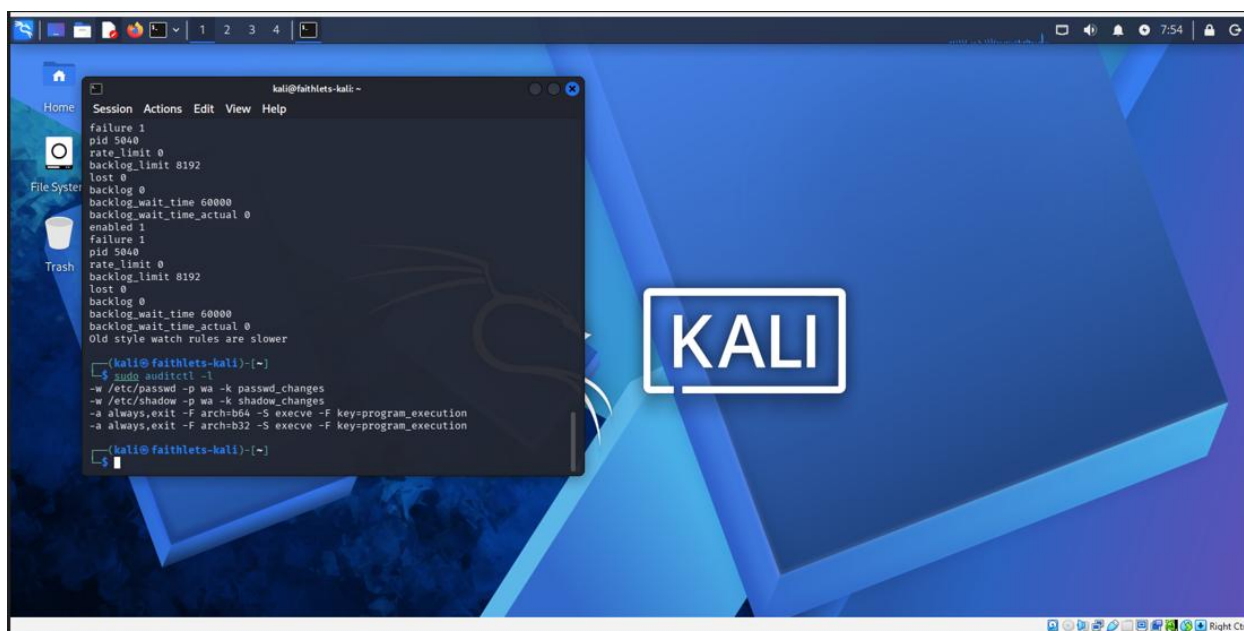


Figure 4: Custom audit rules successfully loaded with auditctl -l

Auditd was successfully installed and started. The service status showed active (running), while auditctl -s confirmed that the Linux audit subsystem was enabled.

Auditd was enabled to start automatically at boot, supporting continued collection of audit records after system restarts.

Custom rules were loaded and verified with auditctl -l. The active rules monitored /etc/passwd, /etc/shadow and program execution by using the keys passwd_changes, shadow_changes and program_execution.

The configuration confirmed that the audit subsystem was ready to capture and retrieve events generated during the practical test.

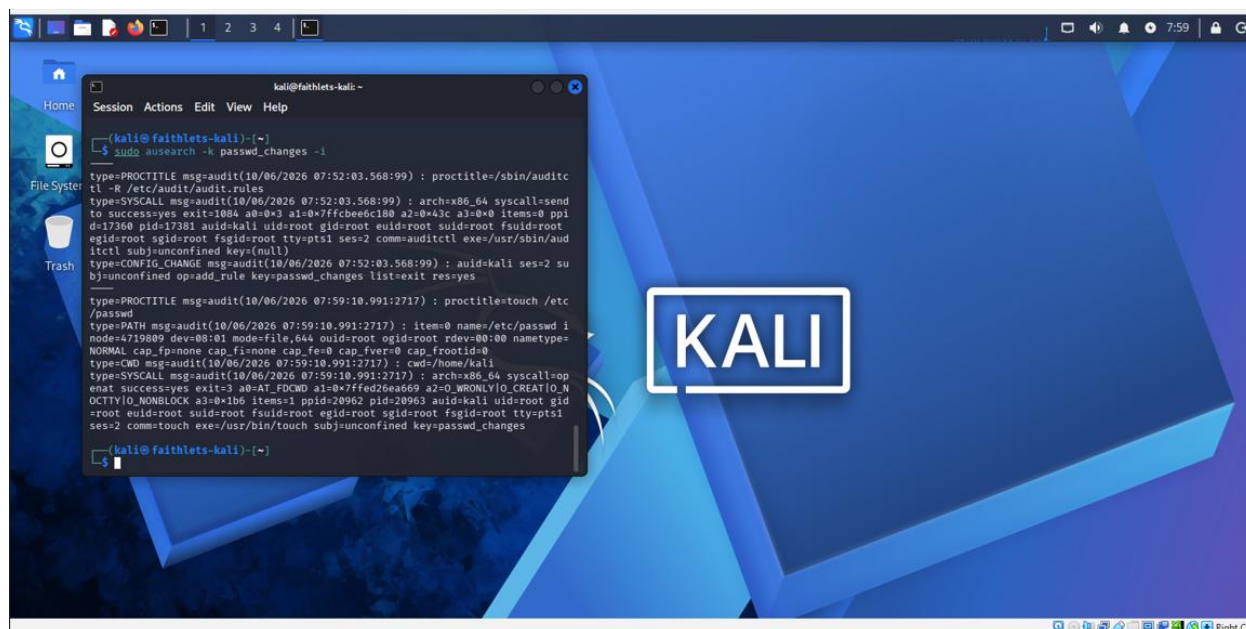


Figure 5: Successful ausearch result for the passwd_changes audit key

A benign metadata event was generated against /etc/passwd and retrieved with ausearch. The event showed proctitle=touch /etc/passwd, name=/etc/passwd, success=yes, comm=touch, exe=/usr/bin/touch and key=passwd_changes. The event was recorded on 6 October 2026 at 07:59:10. Monitoring /etc/passwd is important because unexpected activity involving account files should be traceable and investigated.

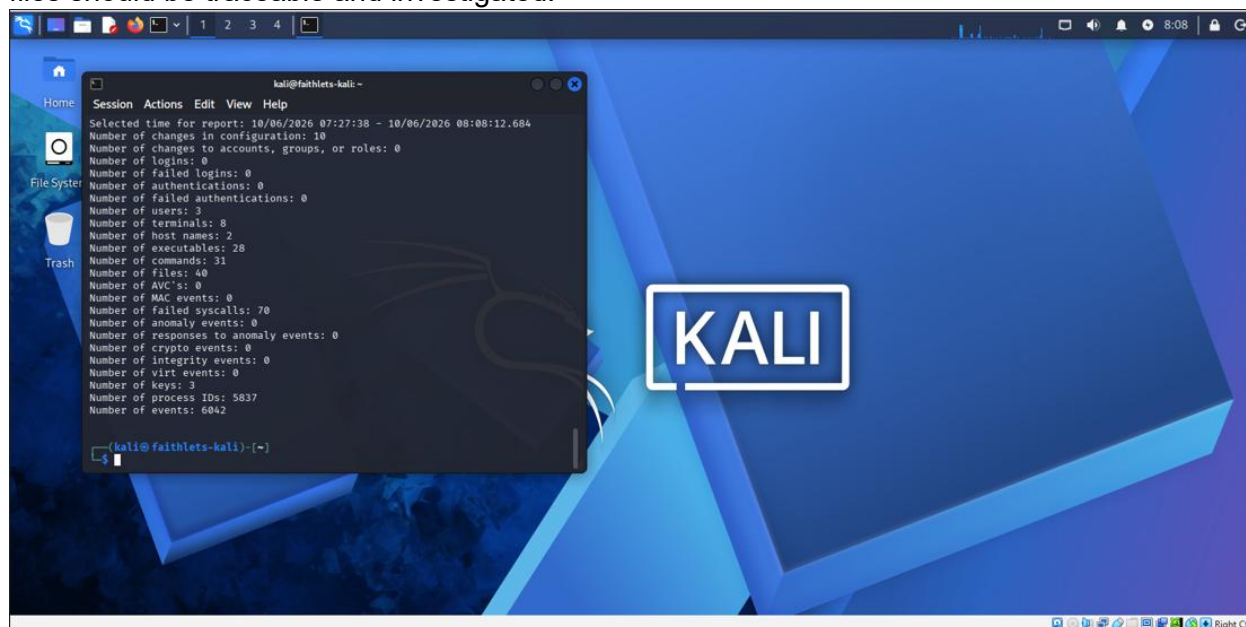


Figure 6: Aureport summary confirming collected audit activity

The aureport summary completed successfully and reported 6,042 total audit events for the reporting period, including commands, executables, file events, configuration changes and failed syscalls. This confirms that audit records were being collected and could be analysed for monitoring and investigation.

Recommendation

Ensure audit logging services generate and retain audit records to support traceability, monitoring, and forensic investigations.

System journal data was reviewed to identify normal system activity, privilege-use activity, authentication events, warnings and errors. The outputs contained timestamps, service or process information and the hostname faithlets-kali, providing traceable evidence for security monitoring.

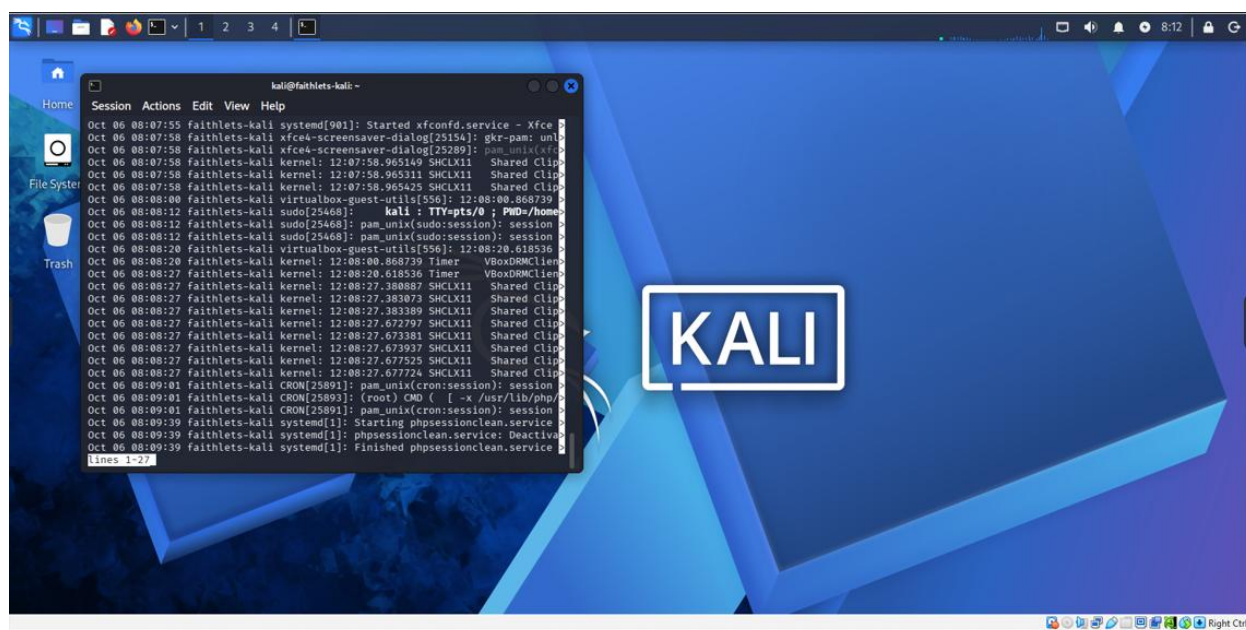


Figure 7: Recent system events returned by journalctl

The journal contained current entries from systemd, sudo, kernel, CRON and other services, confirming that central journal logging was operating.

Objective

To review available Linux logs and identify evidence of monitoring and security-related activities.

Observations

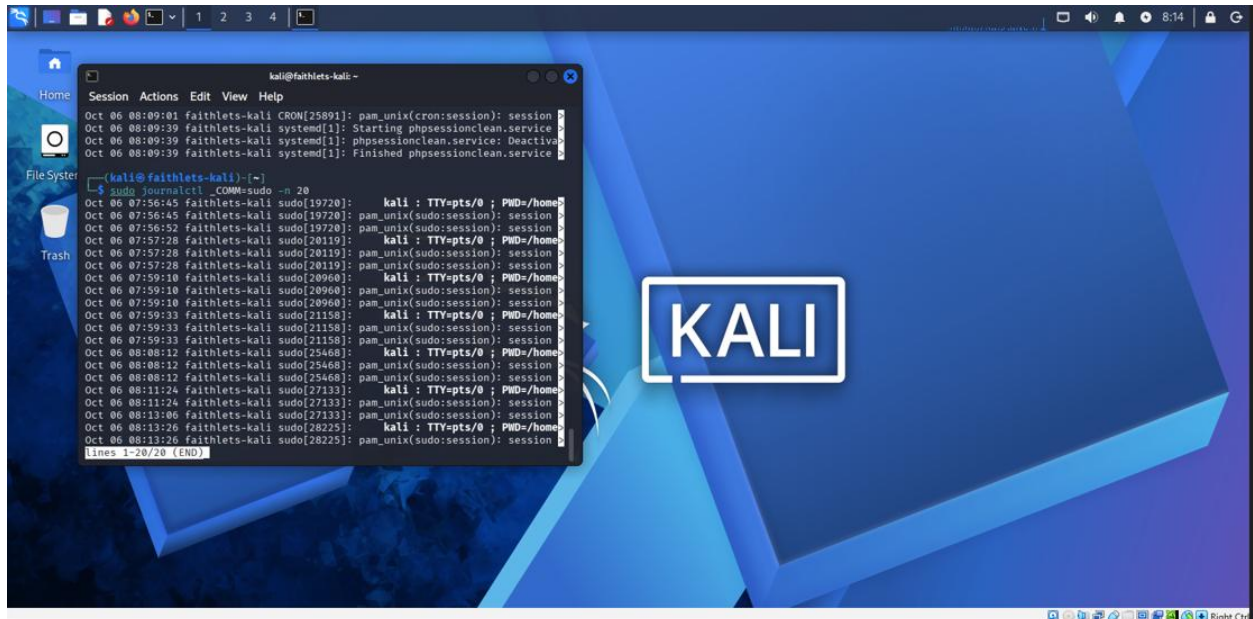


Figure 8: Sudo privilege-use activity

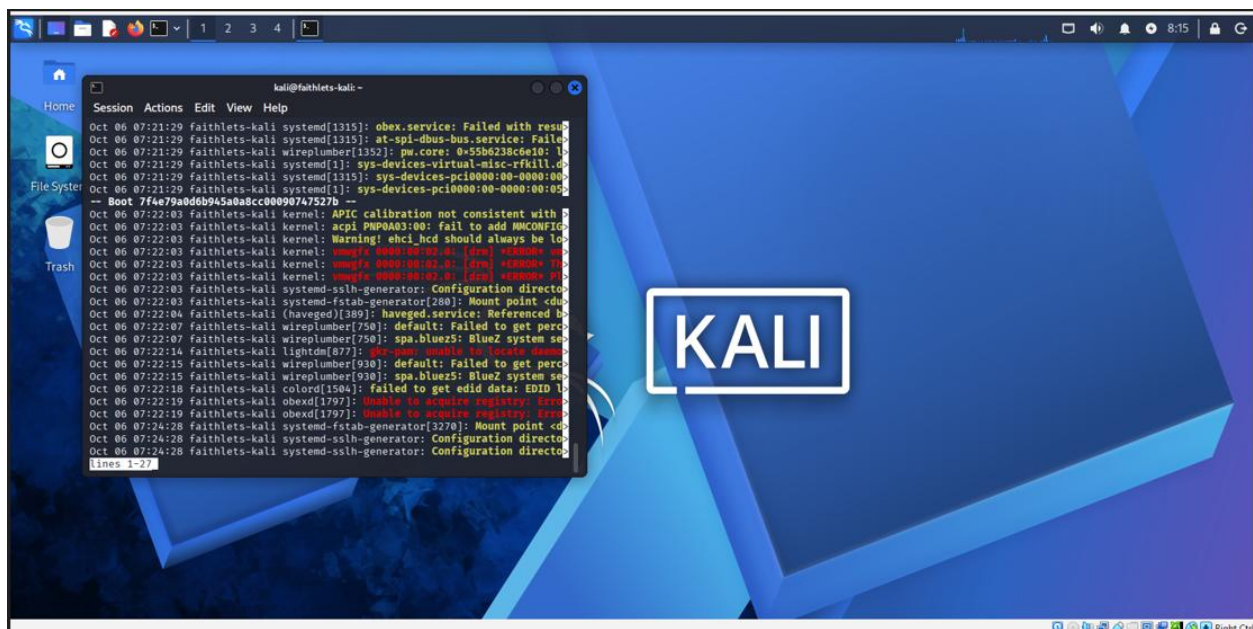


Figure 9: Journal warnings and errors

Filtered journal output showed sudo and pam_unix(sudo:session) events for the kali account, providing evidence of privileged command and session activity.

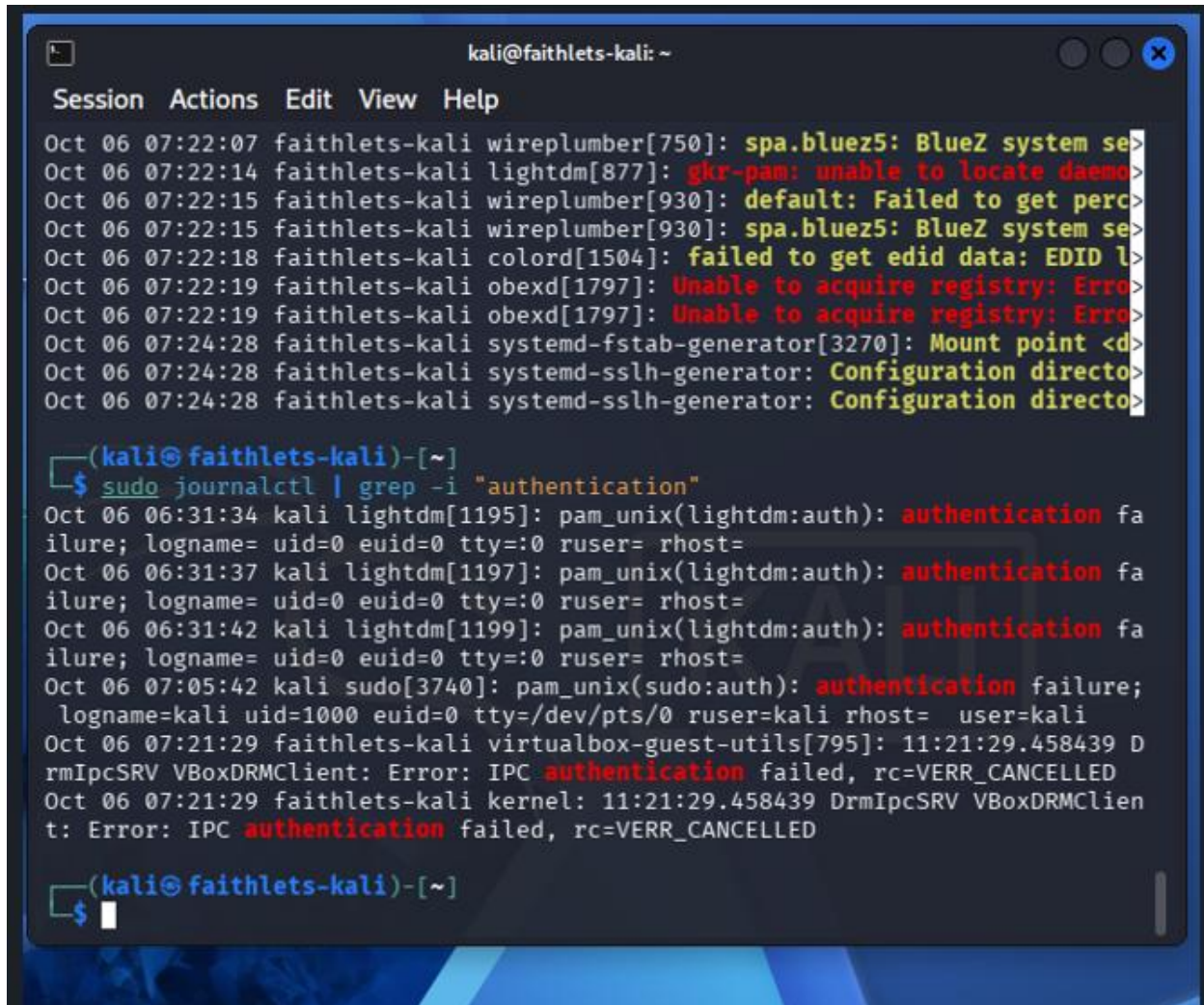
The following command was executed:

```
journalctl --since "today"
```

The result showed:

Priority-filtered journal output returned warning and error events from the operating system and services. These records support operational investigation and troubleshooting.

Review of the system log directories showed:

A screenshot of a terminal window titled 'kali@faithlets-kali: ~'. The terminal shows a list of system logs with timestamps and messages. The messages include various system events, some of which are highlighted in red to indicate errors or warnings. The logs show authentication failures for 'pam_unix' and 'sudo'. The terminal also shows a menu bar with 'Session', 'Actions', 'Edit', 'View', and 'Help'. The logs are filtered by the command 'journalctl | grep -i "authentication"'. The output shows several lines of log entries, including 'authentication failure' and 'authentication failed' messages. The terminal window has a dark background and a light blue border.

```
kali@faithlets-kali: ~  
Session Actions Edit View Help  
Oct 06 07:22:07 faithlets-kali wireplumber[750]: spa.bluez5: BlueZ system se>  
Oct 06 07:22:14 faithlets-kali lightdm[877]: gkr-pam: unable to locate daemo>  
Oct 06 07:22:15 faithlets-kali wireplumber[930]: default: Failed to get perc>  
Oct 06 07:22:15 faithlets-kali wireplumber[930]: spa.bluez5: BlueZ system se>  
Oct 06 07:22:18 faithlets-kali colord[1504]: failed to get edid data: EDID l>  
Oct 06 07:22:19 faithlets-kali obexd[1797]: Unable to acquire registry: Erro>  
Oct 06 07:22:19 faithlets-kali obexd[1797]: Unable to acquire registry: Erro>  
Oct 06 07:24:28 faithlets-kali systemd-fstab-generator[3270]: Mount point <d>  
Oct 06 07:24:28 faithlets-kali systemd-sshd-generator: Configuration directo>  
Oct 06 07:24:28 faithlets-kali systemd-sshd-generator: Configuration directo>  
  
(kali@faithlets-kali)-[~]  
$ sudo journalctl | grep -i "authentication"  
Oct 06 06:31:34 kali lightdm[1195]: pam_unix(lightdm:auth): authentication fa  
ilure; logname= uid=0 euid=0 tty=:0 ruser= rhost=  
Oct 06 06:31:37 kali lightdm[1197]: pam_unix(lightdm:auth): authentication fa  
ilure; logname= uid=0 euid=0 tty=:0 ruser= rhost=  
Oct 06 06:31:42 kali lightdm[1199]: pam_unix(lightdm:auth): authentication fa  
ilure; logname= uid=0 euid=0 tty=:0 ruser= rhost=  
Oct 06 07:05:42 kali sudo[3740]: pam_unix(sudo:auth): authentication failure;  
logname=kali uid=1000 euid=0 tty=/dev/pts/0 ruser=kali rhost= user=kali  
Oct 06 07:21:29 faithlets-kali virtualbox-guest-utils[795]: 11:21:29.458439 D  
rmIpcSRV VBoxDRMClient: Error: IPC authentication failed, rc=VERR_CANCELLED  
Oct 06 07:21:29 faithlets-kali kernel: 11:21:29.458439 DrmIpcSRV VBoxDRMClie  
nt: Error: IPC authentication failed, rc=VERR_CANCELLED  
  
(kali@faithlets-kali)-[~]  
$
```

Figure 10: Authentication failure events

The authentication review identified pam_unix authentication failures, including a sudo authentication failure associated with the kali account. Repeated or unexplained authentication failures should be investigated because they can indicate user error, credential problems or attempted unauthorised access.

Control Owner
Linux Administrator



The screenshot shows a Kali Linux desktop with a terminal window open. The terminal displays the output of a Lynis audit, including the version (3.1.6), a description of the tool, and the results of a specific audit command. A large 'KALI' logo is prominently displayed on the right side of the screen.

```
kali@faithlets-kali: ~
Session Actions Edit View Help

Please submit a pull request on GitHub to include your OS version and the e
nd date of this OS version is being supported
URL: https://github.com/CISOfy/lynis

Lynis 3.1.6

Auditing, system hardening, and compliance for UNIX-based systems
(Linux, macOS, BSD, and others)

2007-2025, CISOfy - https://cisofy.com/lynis/
Enterprise support available (compliance, plugins, interface and tools)

[Tip]: Enhance lynis audits by adding your settings to custom.prfl (see /etc
/lynis/default.prfl for all settings)

(kali@faithlets-kali): [~]
$ sudo grep -E "hardening_index|tests_performed" /var/log/lynis-report.dat
hardening_index=61

(kali@faithlets-kali): [~]
$
```

Figure 12: Lynis Hardening Index 61

The Lynis audit completed successfully on faithlets-kali. Lynis version 3.1.6 produced a Hardening Index of 61, which provides a baseline for measuring future hardening improvements.

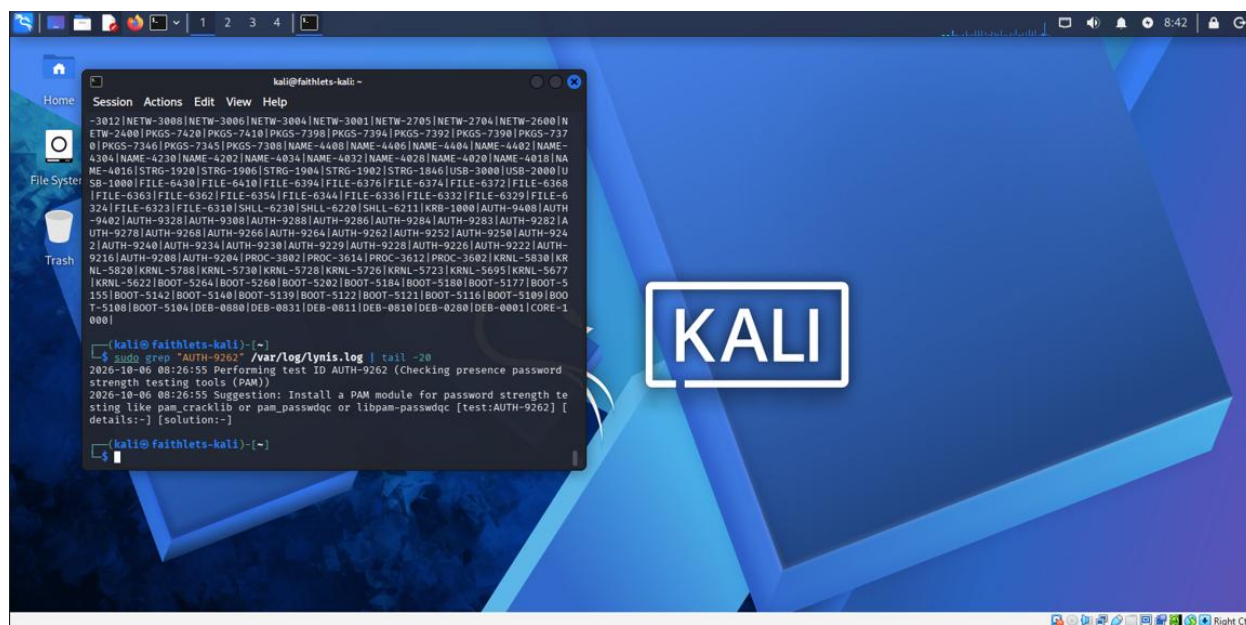


Figure 13: Lynis Finding AUTH-9262 – Password-strength testing

Lynis identified that a PAM password-strength testing module was not present and recommended installing a suitable password-strength module.



Figure 14: Lynis Finding AUTH-9282 – Password expiry

Lynis identified password-protected accounts without expiry dates and recommended setting expiry dates where appropriate.

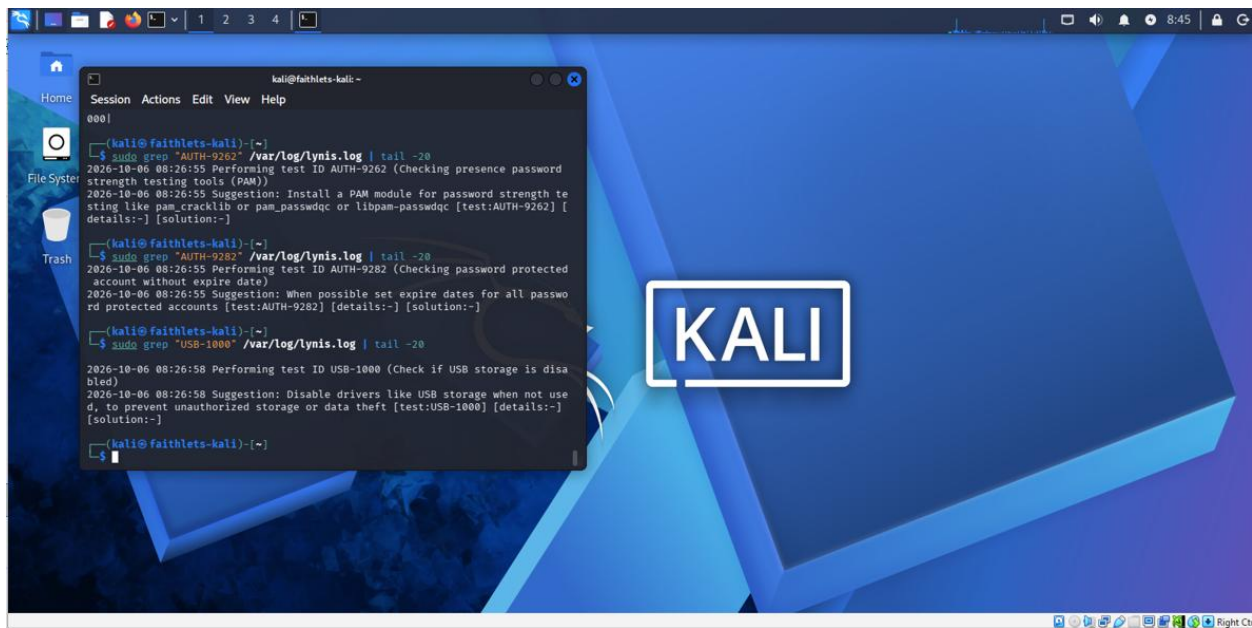


Figure 15: Lynis Finding USB-1000 – USB storage

Lynis recommended disabling USB-storage drivers when removable storage is not required to reduce the risk of unauthorised storage or data theft.

Security Measure	Result
Lynis Version	3.1.6
Hardening Index	61
Tests Performed	232
Plugins Enabled	1
Firewall	Enabled
Malware Scanner	Not Installed

Module 3: Lynis Findings and Risk Interpretation

Finding	Risk / Control Issue	Evidence	Likely Owner	Recommended Remediation	Priority
AUTH-9262: Password strength control	No PAM password-strength module was identified. Weak passwords could therefore be accepted, increasing the risk of account compromise.	Lynis identified AUTH-9262 and recommended installing a PAM module for password-strength testing.	Linux/System Administrator	Install and configure an appropriate PAM password-quality module and align the settings with the approved password policy.	High
AUTH-9282: Password expiry	Password-protected accounts were identified without expiry dates. Credentials that remain unchanged for long periods may increase exposure if a password is compromised.	Lynis identified AUTH-9282 and recommended setting expiry dates for password-protected accounts where appropriate.	Linux/System Administrator	Review password-aging requirements and configure suitable expiry settings in line with organisational policy.	Medium
USB-1000: USB storage	USB storage remains available. Where removable media is not required, this may increase the risk of unauthorised data transfer or introduction of unapproved files.	Lynis identified USB-1000 and recommended disabling USB-storage drivers when they are not required.	Linux/System Administrator / Security Team	Disable USB-storage functionality where there is no business need. Where removable media is required, manage access through an approved removable-media policy.	Medium

6. Control Monitoring and Governance Mapping

Control Monitoring Table

Control / Objective	Evidence	Owner	Observed Status	KPI/KRI or Threshold	Risk / Significance	Remediation	Retest / Follow-up
Audit Event Monitoring	Auditd active; rules loaded; ausearch and aureport returned records	Linux Administrator	Operating	Successful retrieval of configured audit-key events; zero unexpected logging gaps	High if audit collection becomes unavailable	Maintain auditd/rules and log retention	Generate benign event and confirm ausearch/aureport after changes
Authentication Monitoring	journalctl showed sudo privilege activity and pam_unix authentication failures	Linux Administrator	Operating	Authentication and privilege-use events available; investigate repeated failures above approved threshold	High if failures are unexplained or logging becomes unavailable	Maintain journal collection and investigate repeated failures	Repeat controlled sudo/authentication test and confirm journal evidence
Vulnerability Management	Lynis warning: "Found one or more vulnerable packages"	Infrastructure Administrator	Deficient	Zero critical vulnerabilities and timely patching of systems	High. Vulnerable software may be exploited if not updated	Apply operating system and package updates	Run apt update/upgrade and rerun Lynis to verify warning is cleared
Password Management	Lynis AUTH-9262 and AUTH-9282	Linux/System Administrator	Needs Improvement	Password-strength and aging settings meet approved policy	High for missing strength validation; Medium for expiry gap	Configure PAM password quality and approved aging settings	Rerun Lynis after remediation and verify settings

Control / Objective	Evidence	Owner	Observed Status	KPI/KRI or Threshold	Risk / Significance	Remediation	Retest / Follow-up
File Integrity Monitoring	Lynis recommended installation of a file-integrity monitoring tool	Security Administrator	Deficient	Critical file changes should be monitored and reported	High. Unauthorized modifications may go undetected	Implement AIDE, OSSEC, or another file-integrity solution	Verify monitoring reports and confirm file-change alerts are generated
Malware Protection	Lynis result showed "Malware Scanner [X]"	Security Administrator	Needs Improvement	Malware protection should be installed and active	Medium. Malware infections may not be detected promptly	Install and maintain a malware-scanning solution	Rerun Lynis and confirm malware scanner is detected
Kernel Logging	Lynis warning: "klogd is not running"	Linux Administrator	Needs Improvement	Kernel events should be logged and available for review	High. Important kernel-level events may be missed	Enable kernel logging services	Review kernel logs and verify logging functionality
Secure Configuration Management	Lynis identified multiple hardening recommendations including file permissions, legal banner, and system-hardening settings	Linux Administrator	Needs Improvement	Continuous improvement of system-hardening score	Medium. Weak configurations increase the attack surface	Implement recommended system-hardening measures	Rerun Lynis and compare hardening score with baseline (61)

7. Governance Escalation Analysis

Highest-Priority Finding

The highest-priority selected Lynis finding is AUTH-9262, the absence of PAM password-strength testing. Weak password controls can increase the likelihood that weak passwords are accepted. This finding should be addressed first, followed by password-expiry and removable-media hardening based on organisational requirements.

Ownership of Findings

Findings	Owner
Audit Logging and Monitoring	Linux Administrator
Authentication Monitoring	Linux Administrator
Vulnerable packages	Infrastructure Administrator
Weak password controls	Identity Administrator
Missing file integrity monitoring	Security Administrator
Malware protection gap	Security Administrator

Escalation Thresholds

Escalation should occur when:

- Audit logs remain unavailable for more than 24 hours.
- Authentication logs are unavailable.
- High-risk vulnerabilities remain unresolved.
- Critical controls fail retesting.
- Security-hardening recommendations remain unaddressed.

Evidence of Successful Remediation

Successful remediation would be demonstrated through:

- Available audit and authentication logs
- Successful ausearch and aureport results
- Improved Lynis hardening score
- Removal of vulnerable-package warnings
- Presence of file-integrity monitoring
- Presence of malware-scanning capability

8. SIEM and Continuous Monitoring Mapping

Data Source	Monitoring Purpose	SIEM Collection	Escalation Priority
Auditd	Audit-event monitoring	Yes	High
ausearch	Audit investigations	Yes	High
aureport	Audit reporting	Yes	Medium
journalctl	System monitoring	Yes Yes	Medium
Authentication Logs	Login monitoring	Yes	High
Lynis Reports	Security posture monitoring	Yes	Medium

9. Remediation and Retest Plan

Findings	Recommended Action	Owner	Retest Method
Vulnerable Packages	Apply security updates	Infrastructure Administrator	Re-run Lynis
Audit Logging and Monitoring	Maintain audit logging and configured audit rules	Linux Administrator	Generate a benign test event and verify ausearch output
Authentication Monitoring	Maintain authentication logging and review repeated failures	Linux Administrator	Generate a controlled authentication event and confirm journal output
Weak Password Controls	Configure password policies	Identity Administrator	Verify configuration settings
Missing File Integrity Monitoring	Install AIDE or OSSEC	Security Administrator	Verify monitoring reports
Malware Protection Gap	Install malware scanner	Security Administrator	Verify Lynis detection
Kernel Logging Weakness	Enable kernel logging	Linux Administrator	Review kernel log output

10. Conclusion

The assessment confirmed that the Linux monitoring and auditing controls were functioning in the authorised Kali Linux lab. Auditd was active, the audit subsystem was enabled, the required custom audit rules were loaded, and a benign /etc/passwd event was successfully retrieved with ausearch. Aureport also generated a valid audit summary.

The journal review produced current system events, sudo privilege-use records, authentication failures, and warning/error events. These records provide useful evidence for operational monitoring, incident investigation and control assurance.

The Lynis assessment used version 3.1.6 and established a Hardening Index baseline of 61.

The selected findings were AUTH-9262 for password-strength testing, AUTH-9282 for password expiry, and USB-1000 for removable-media exposure. These findings should be remediated according to approved organisational policy and then retested.

Overall, the assessment demonstrates a functioning monitoring baseline while identifying opportunities to strengthen authentication and removable-media controls. Ongoing monitoring, assigned ownership, defined escalation thresholds and formal retesting will improve control assurance over time.

Appendix A: Evidence Register

Evidence captured during the Linux security monitoring and auditing practical

Figure	Evidence captured	Command / reference
Figure 1	Auditd service active and running	sudo systemctl status auditd
Figure 2	Audit subsystem enabled	sudo auditctl -s (enabled=1)
Figure 3	Auditd enabled at boot	sudo systemctl enable auditd
Figure 4	Custom audit rules loaded	sudo auditctl -l
Figure 5	passwd_changes audit event retrieved	sudo ausearch -k passwd_changes -i
Figure 6	Audit activity summary	aureport summary
Figure 7	Recent system events	journalctl recent system events
Figure 8	Sudo privilege-use activity	sudo journalctl _COMM=sudo -n 20
Figure 9	Journal warnings and errors	journalctl warning/error review
Figure 10	Authentication failure events	journalctl authentication review
Figure 11	Lynis version 3.1.6	lynis show version
Figure 12	Lynis Hardening Index 61	Lynis report hardening_index=61
Figure 13	Lynis password-strength finding	AUTH-9262
Figure 14	Lynis password-expiry finding	AUTH-9282
Figure 15	Lynis USB-storage finding	USB-1000